

Inhibition of the growth of premalignant and malignant human oral cell lines by extracts and components of black raspberries.



Black raspberries are a rich natural source of chemopreventive phytochemicals. Recent studies have shown that freeze-dried black raspberries inhibit the development of oral, esophageal, and colon cancer in rodents, and extracts of black raspberries inhibit benzo(a)pyrene-induced cell transformation of hamster embryo fibroblasts. However, the molecular mechanisms and the active components responsible for black raspberry chemoprevention are unclear. In this study, we found that 2 major chemopreventive components of black raspberries, ferulic acid and beta-sitosterol, and a fraction eluted with ethanol (RO-ET) during silica column chromatography of the organic extract of freeze-dried black raspberries inhibit the growth of premalignant and malignant but not normal human oral epithelial cell lines. Another fraction eluted with CH₂Cl₂/ethanol (DM:ET) and ellagic acid inhibited the growth of normal as well as premalignant and malignant human oral cell lines. We investigated the molecular mechanisms by which ferulic acid and beta-sitosterol and the RO-ET fraction selectively inhibited the growth of premalignant and malignant oral cells using flow cytometry and Western blotting of cell cycle regulatory proteins. There was no discernable change in the cell cycle distribution following treatment of cells with the RO-ET fraction. Premalignant and malignant cells redistributed to the G₂/M phase of the cell cycle following incubation with ferulic acid. beta-sitosterol treated premalignant and malignant cells accumulated in the G₀/G₁ and G₂/M phases, respectively. The RO-ET fraction reduced the levels of cyclin A and cell division cycle gene 2 (cdc2) in premalignant cells and cyclin B1, cyclin D1, and cdc2 in the malignant cell lines. This fraction also elevated the levels of p21waf1/cip1 in the malignant cell line. Ferulic acid treatment led to increased levels of cyclin B1 and cdc2 in both cell lines, and p21waf1/cip1 was induced in the malignant cell line. beta-sitosterol reduced the levels of cyclin B1 and cdc2 while increasing p21waf1/cip1 in both the premalignant and malignant cell lines. These results show for the first time that the growth inhibitory effects of black raspberries on premalignant and malignant human oral cells may reside in specific components that target aberrant signaling pathways regulating cell cycle progression.